THE PRACTICE OF STATISTICS

The Nature of Statistical Consulting

Recall: Statistical Consulting

- Interpersonal interaction
- Goal: increasing the utility of information
- Disseminated within and among disciplines
- Uses STATISTICS

Who Are the Clients?

- Other Researchers (e.g., Academe)
  - Research design
  - Experiments [design, analysis, strategies like blocking and use of covariates]
  - Surveys [including questionnaire design, sampling design, sample size determination]
  - Data Analysis
  - Modeling
  - Computing

Who Are the Clients?

- Business/Corporate
  - Traditional MR vs. Data mining
  - Forecasting (for planning, sales targeting, supply chain management)
  - Business analytics (e.g., for CRM, Customer Care, Revenue Assurance)
  - Consumer metrics (e.g., lifetime valuation, risk)
  - Segmentation and tailor-fitting of marketing strategies/CRM

Who Are the Clients?

- Industry/Manufacturing
  - Product design/development/improvement
    - Design of experiments
    - Optimal designs
  - Quality monitoring
    - Quality control (control charts)
    - Process capability analysis
    - Cause and effect analysis
  - Reliability analysis
    - Probability modeling
    - Lifetime analysis

Who Are the Clients?

- Organizational development
  - Client/Customer satisfaction (including scale development)
  - Peer evaluation
- Banks/Financial Institutions
  - Risk modeling/management
  - Credit scoring
  - Fraud detection
  - Customer valuation

Who Are the Clients?

- Public Sector/Multilateral Agencies
  - Official Statistics/Survey Sampling/Estimation
  - Impact Assessment
    - Sampling
    - Quasi-experiments
    - Modeling
      - Propensity score matching
      - Switching regressions
      - Counterfactual simulations
  - Monitoring and Evaluation Designs
  - Policy Formulation/Policy Analysis
  - Benchmarking/Road Map

Desirable Attributes

- Interpersonal Skills
  - Listening to the client
  - Communication: increase chance that client & consultant understand each other
  - Team player

Desirable Attributes

- Technical Skills: ***Statistical thinking***
  - Data collection and presentation
  - Statistical inference
  - Regression analysis
  - Experimental designs
  - Sample surveys
  - Multivariate analysis
  - Time series analysis
  - Statistical packages
  - Coding

Desirable Attributes

- Self-management skills
  - Competence in time management
  - Shifting of focus (simultaneous projects)
- Business management skills
- Personal qualities
  - Interest
  - Personal appearance
  - Ethics
  - Curiosity

Desirable Attributes

- Have a genuine desire to solve real-life problems and help others solve problems
- Be able to help investigators (client) formulate their problem in quantifiable terms
- Be able to listen carefully and to ask probing questions
- Have a broad knowledge and true understanding of statistical and scientific methods

Effective Statistical Consultant Should…

- **Be able to adapt existing statistical procedures to novel environmen**ts
- Be able to locate or develop good statistical procedures in a timely fashion
- Be able to keep abreast of developments in statistics (familiar with the literature)
- Be willing to meet deadlines, even if it requires substantial extra effort

Effective Statistical Consultant Should…

- Be able to understand something about the clients' subject matter and speak a bit of the clients' language
- Be a good teacher-most success in consulting depends on being able to help others understand statistical tools, and their strengths and weaknesses
- Be willing to settle for a reasonably correct approximate solution, then go on to the next problem

Effective Statistical Consultant Should…

- Be able to identify important problems (and thus avoid spending too much time on projects of little significance)
- Have confidence to use as simple a procedure as will get the job done (design or analysis)
- Be able to convince others of the validity of a solid solution and see to it that proper action is taken

Effective Statistical Consultant Should…

- Be a good problem solver
- Be willing to meet clients regularly on their home ground, and take the responsibility to meet and communicate with all members of the working team
- Be diplomatic and know when to bend, when to stand firm, and how to help smooth conflicts among other team members.

Effective Statistical Consultant Should…

- Be willing to take the time to check and double-check procedures and results
- Be able to communicate effectively (writing and orally)
- Be able to make a good estimate of how much effort will be required to solve the problem without actually having to solve the problem itself

On Competence

- There are plenty of jobs for statistical consultants who fall short of the ideal.
- You can’t know everything
- You’re going to make mistakes

Summary…Important Values

- Patience
- Timeliness
- Humility
- Openness to New Ideas
- Eagerness to Learn
- Respect and Courtesy

Reminders…

- Respect for client and his/her discipline
- Statistical/Professional Integrity
  - Ethics (see PSAI, ISI Code of Ethics)
- Feasibility of the Workplan/Timelines
- Intellectual Rights/Plagiarism/NDA
- Constant Checking of the Budget
- Ask to see laboratory, apparatus, field trials, study site/area

The Work of Consulting

- Consulting can develop you as a statistician. It offers you an opportunity to grow and is very challenging.
- You can find social satisfaction in consulting and form lasting friendships.
- Good consulting helps improve the world.
- Consulting can be exciting, especially when you put together information from diverse sources to solve a thorny problem.
- You can learn about new developments in many fields from your clients.

Client and Peer Expectations

- Your clients want you to be technically competent, interested in helping them solve their problems, and able to relate to them as individuals.
- Your clients want you to be a good business manager, able to manage time and money.
- The Statistical Community wants you to Advance the Knowledge and Prestige of Statistics (ADVOCACY).
- The statistical community wants you to resist client pressures to compromise “too much”.
- The statistical community is increasing its emphasis on statistical consulting.

Choosing a Technique

- Almost all clients want you to use techniques already accepted in their fields. In using a new technique, you are taking a cause. Make sure it is worthy to your client.
- It’s hard to make and meet deadlines when you use a technique new to you.
- Use the simplest technique that will do the job, not the fanciest you can handle.
- If the client’s solution is decent, don’t push your own unless it is much better. This strategy will most likely give you and ideal client in the long run.

Choosing a Technique

- Be ready to help clients design data collection systems, including questionnaires
- Work very hard for client understanding and acceptance of your ideas in the design.
- Agree how much of the work the client must do and/or understand before choosing a technique.
- Get your purpose and intentions aligned with the client's. Be 100% responsible for getting your part of the job done. Let the client know if you cannot be counted on to participate in the study at the earliest stage.

Basic Rules (Techniques)

- Simple Over Sophisticated Methods
- Practicality Over Statistical Rigidity
- Speak in the language of the client

The Necessary Marriage of Practice and Research

- Not all real-life problems have statistical solutions.
- Feedback from statistical practice to research.
- Tools needed in statistical practice can be very specific to the problem.
- Statistical theory is dynamic.

End of Topic “Stat Consulting Practice”